

Facilitator guide

A random sample from the mathematics atlas - September 2026

Give one student page at a time. Let learners try their own examples before offering a hint or a new representation. These are reviewed worksheet trials; classroom pacing is still untested.

INVESTIGATION	PAGE
GA-26 Can colors detect a knot? Three colors; careful tracing	2
GA-29 The disappearing strip Fractions and powers; optional infinite arguments	7
GA-11 What must an addition machine reveal? Signed fractions and algebra; later, irrational inputs	11
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Preparation, staged hints, all solutions and source notes follow in the same order. Calculus is a real prerequisite for AP-07, rather than an optional label.

Can three colors catch a knot?

Facilitator preparation, staged hints and complete solutions

What counts as a satisfying result

Solve A's coloring and prove B cannot have a nonconstant one by forced propagation. Full knot non-equivalence is a later stop after local checks plus the supplied theorem.

Promising and substantial, with a real visual-prerequisite risk. Over/under tracing and coloring an arc rather than each visible stroke are genuinely hard. Page 1 must succeed before continuing. Pages 3-4 may be a second meeting. Do not promise that a seven-year-old's rainbow picture alone proves the trefoil is knotted.

Prerequisites

Entry: Track a continuous line through over/under crossings; use three distinct symbols or colors. No arithmetic required.

Core explanation: Apply a local same-or-all-different rule and organize cases; distinguish "cannot find" from an impossibility reason. Spatial tracing is a real prerequisite.

Extension: Use five color-equality patterns to check a local move and reason about reversible one-to-one correspondences. The general Reidemeister theorem is an explicitly supplied fact, not proved by coloring.

Materials, preparation and pace

three pencils or labels R/B/G; one loose overhand knot with its ends taped together and one plain closed loop of cord per pair, optional but recommended; paper diagrams large enough to trace; a spare open cord for making local moves. Preparation: about 15 minutes.

Tie a loose overhand knot before joining the ends to make a trefoil. Tape the ends of a second unknotted cord to make a plain loop. Do not ask a child to test unknotting by unsealing joins. For matching the braid drawing, two open strands can be crossed three times and then the same-side bottom/top ends joined; the resulting closed braid has one component. A failed untying attempt is evidence for a conjecture, not a proof.

0-10 play with closed cords and trace crossings; 10-15 state coloring rule; 15-35 investigate A and B; 35-40 whole-body over/under trace; 40-55 choose two-move checks OR three-strand extension; 55-60 state exactly what was proved. The complete local-invariance proof may require a second session.

New knot object and invariant. Week 2 has invariants, but no crossing/arc convention or topology. The atlas's assumed standard trefoil has been replaced by explicit, independently verified geometry.

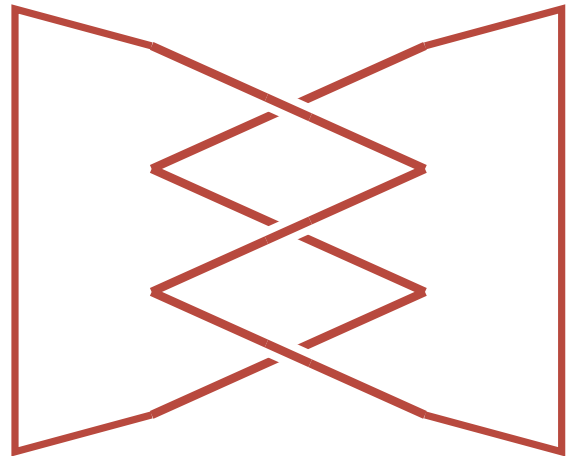
Can three colors catch a knot? - solutions

Prompt numbers match the student pages; back-pocket questions are noted.

A: one of six nonconstant colorings



B: only constant colorings



Color key: R = red, B = blue, G = green. The letters in the tables give the same information in monochrome.

26.1

Only the MIDDLE crossing changes which strand is on top. Both diagrams have the same planar shadow and exactly 3 crossings. Following either diagram traverses both braid strands and both outer closures before returning, so each has one component. A is the closure of σ_1^3 , a trefoil; B is the closure of $\sigma_1\sigma_1^{-1}\sigma_1$.

26.2

A cannot become a circle; the later coloring certificate proves it. B can: cancel its first two opposite crossings by a type II move, leaving a one-crossing closed braid, then remove its curl by type I. Physical difficulty or matching coloring counts alone would not prove this.

26.3

A has 3 arcs/sections between successive underpasses. Each arc can contain an overpass. Starting at the left closure, its top-left portion is arc A; top-right portion arc B; the left strand just below first crossing arc C. Because the marked dot may lie inside a section, returning to that section is not a fourth arc.

Can three colors catch a knot? - solutions

26.4

Given overcolor and one undercolor, the other undercolor is uniquely forced: same if they agree, otherwise the third color. Thus valid examples are RRR and RBG; RRB is invalid. The gap is an underpass of a single cord, not a cut end; color is permitted to change there because diagram arcs terminate at undercrossings.

26.5

A: start top colors R, B. From top to bottom the color pairs are (R, B), (G, R), (B, G), (R, B). Both outer joins agree, and every crossing contains all 3 colors. B with the same start gives (R, B), (G, R), (R, B), (G, R), which fails the outer joins. Any two different top colors behave by renaming in the same way; equal colors stay equal. Hence B has no nonconstant coloring.

26.6

For B the bottom pair differs from the top pair when the two top colors are different. Since each bottom endpoint joins the SAME-side top endpoint, the drawing cannot be colored consistently. The local rule forces every intermediate color; there is no alternative hidden choice.

26.7

A has 9 total colorings: 3 equal top pairs give constant colorings, and 6 unequal ordered top pairs give nonconstant ones. B has 3 total, all constant. A plain circle has one arc and therefore 3 constant colorings. Global fixed drawing/colors are counted; color permutations are distinct here. On later move-check pages, renaming is only a way to reduce proof cases.

26.8

The curl must remain entirely red. Its overpass and the entering undersection are already the same color, so the outgoing undersection is forced red. This proves a unique extension for any single starting color; reversing the move forgets no choice.

26.9

For input (R, B), the pair between crossings is (G, R) and after both crossings returns to (R, B). For (R, R), every section is R. The corresponding straight strands keep their input colors. Both operations are uniquely reversible. Do not accidentally draw two crossings of the SAME sign; those are not removable by this move.

26.10

A pair of colors is either equal or different. Any equal pair is a renaming of RR and any unequal ordered pair is a renaming of RB. The rule depends only on equality/difference, so those two trials cover all 9 assignments. Each crossing uniquely determines the new undercolor.

Can three colors catch a knot? - solutions

26.11

Yes: B's first two crossings are positive then negative, with the same physical strand passing over both; they cancel. The remaining one-crossing closure is a curl of the unknot. Changing one crossing would require a strand to pass through another and is not one of the legal deformations.

26.12

Both diagrams send the following top colors to the same bottom colors, left to right: RRR→RRR; RRB→BRR; RBR→BGR; RBG→RGR; BRR→GGB. Exact 27-case check confirms all. These are ENDPOINT colors, not three arc names. The intermediate colors may differ between drawings; the endpoint correspondence is identical.

26.13

The patterns are: all same; left=middle only; left=right only; middle=right only; all different. Every triple belongs to exactly one type. Color renaming transfers the checked example to every member of its type. Uniqueness follows crossing-by-crossing; reverse moves recover the original.

26.14

Each local move gives a bijection between the entire diagrams' legal colorings because outside colors stay fixed and inside completion is unique in both directions. Therefore a finite sequence preserves the number. Ordinary redrawing also preserves it. The supplied Reidemeister theorem turns any possible unknotting into such a sequence. A has 9 but the circle has 3, impossible. The theorem is necessary; checking a handful of moves without it would not cover arbitrary 3 Ddeformations. The five student cases cover the displayed type III representative. The complete facilitator variant check covers every consistent strand-height order; it must be supplied if the children have not proved those variants. See invariance_variants and the 162-case exact check.

26.15

No. Equal invariant values do not establish equivalence; an invariant may fail to distinguish different knots. For B we additionally gave an explicit sequence to a circle. Do not offer an unsourced nontrivial zero-test knot as if checked in this packet.

Offer hints only after exploration

26.5: Try different colors on the two top strands, then each crossing forces the next color.

26.7: There are three choices for the left top color and three for the right. Which choices close consistently?

26.9: The same strand passes over at both crossings. Track it all the way.

Can three colors catch a knot? - solutions

26.13: Three colors can be all equal, all different, or have just one unequal position.

All knot-move variants, not only the pictured braid

The displayed braid slide is one representative. A complete local check must include all allowed height orders, not just three positive crossings. For signs (a, b, c), compare braid words $(\sigma_1^a, \sigma_2^b, \sigma_1^c)$ with $(\sigma_2^c, \sigma_1^b, \sigma_2^a)$. Six sign triples are legal; exclude $(+, -, +)$ and $(-, +, -)$, which would require a cyclic above/below order of the three strands. The exact script verifies all $6 \times 27 = 162$ color assignments. It also checks both orders of opposite-sign type II crossings (18 assignments), and type I forces one color. Turning a strand around does not change the rule: the two undercolors are symmetric. The operation $a * b = 2b - a$ modulo 3 is idempotent, involutory and self-distributive, giving the same argument for all unoriented local variants. These extra checks belong to the facilitator; students may use their invariance conclusion as a supplied fact after investigating the pictured examples.

Sources and checked scope

Louis H. Kauffman, Knots. §II and §III; “Three Coloring a Knot,” Figures 12-14. Local Fox coloring rule, Reidemeister-move invariance and the theorem bridge. Exact braid drawings and finite tables independently checked.

<https://homepages.math.uic.edu/~kauffman/Tots/Knots>

Jonny Evans, 6.03 Braids: the Wirtinger presentation. example identifying closure of σ_1^3 as the trefoil. Confirms the explicit closed-braid diagram chosen here is a trefoil, rather than relying on a familiar icon.

<https://www.jde27.uk/tg/braids03.html>

Exact checks: `plans/atlas/worksheet-trial/geometry-checks.py` and `geometry-checks-results.json`. Coloring is not a complete knot test. No classroom pilot yet.

More pieces. Less path.

Facilitator preparation, staged hints and complete solutions

What counts as a satisfying result

Explain all 16 addresses and solve the round 6 double challenge, while identifying permanently safe endpoints.

Good finite worksheet; the ambitious title of the original atlas card needs an honest advanced gate. The full uncountability result is not a satisfying low-prerequisite endpoint for every learner. The finite first 3 pages are the core. Infinite page 4 is optional and likely another meeting; do not equate enjoying the picture with proving the theorem.

Prerequisites

Entry: Locate labeled whole numbers and split a segment into three equal parts; count and double.

Core explanation: Fractions with denominators 3 and 9 for budget page; follow finite L/R addresses; distinguish endpoints from interiors.

Extension: Infinite sequences, a nested-interval existence fact and “for every proposed list” reasoning. No calculus, but genuine proof maturity. Do not describe the infinite page as elementary merely because the arithmetic is small.

Materials, preparation and pace

paper number-line strips; pencils in two colors; ruler optional; address slips L/R; scissors optional only for separate play strips; use shading on proof sheets to preserve endpoints. Preparation: about 8 minutes.

0-8 fold/play with thirds; 8-13 state remove-open-middle rule; 13-30 finite cutting and addresses; 30-35 movement: walk L/R branches; 35-50 choose budget or infinity extension; 50-60 explain one certificate. Page 4 often deserves a later meeting.

No documented matching worksheet. Binary branching resembles several earlier tree representations, but here the addresses code nested intervals rather than queries or moves. GA-03 has a related countable zero-length example; this extension tests the false converse.

More pieces. Less path. - solutions

Prompt numbers match the student pages; back-pocket questions are noted.

Round 0



Round 1



Round 2



Round 3



Round 4



29.1

Round 1 keeps $[0, 27]$ and $[54, 81]$. Round 2 keeps $[0, 9]$, $[18, 27]$, $[54, 63]$, $[72, 81]$. Gaps are open: 27 and 54 remain, as do 9, 18, 63, 72. Distinguish a mathematical endpoint from the finite width of ink.

29.2

9, 18, 27, 54 are safe forever because they become endpoints and stay endpoints of surviving intervals. 15 is first removed in round 2, from the gap $(9, 18)$. 40 is removed in round 1, from $(27, 54)$. "Not erased yet" alone is not a forever proof.

29.3

In $[0, 9]$, round 3 removes $(3, 6)$; for example 5 survives two rounds then is erased. Other examples: 21 is an endpoint of the gap $(21, 24)$ and is NOT erased, so choose 22 instead. The other new open gaps are $(21, 24)$, $(57, 60)$, $(75, 78)$.

29.4

LLRR= $[8, 9]$; RLRL= $[60, 61]$. Four choices contribute offsets 0 or 54, 0 or 18, 0 or 6, 0 or 2, leaving length 1. Each address is a recursive interval choice, not a base-two coordinate value.

29.5

$[20, 21]$ has address LRLR: left $[0, 27]$, right $[18, 27]$, left $[18, 21]$, right $[20, 21]$. Partner puzzles may use any of the 16 intervals in the checked address table.

More pieces. Less path. - solutions

29.6

LRLL, LRLR, LRRL, LRRR. After fixed prefix LR, the two remaining positions each have two choices, exhausted by LL, LR, RL, RR. A small binary tree is a completeness certificate.

29.7

There are $2 \times 2 \times 2 \times 2 = 16$ pieces. The first differing letter sends two addresses into left and right thirds of one parent interval, separated by a positive gap, so the final pieces cannot coincide. This finite address bijection proves the count.

29.8

For rounds 0-6: piece counts 1, 2, 4, 8, 16, 32, 64; individual lengths 81, 27, 9, 3, 1, $1/3$, $1/9$; totals 81, 54, 36, 24, 16, $32/3$, $64/9$. Powers notation is optional.

29.9

First win is round 6: $64 > 50$ and $64/9 < 10$. Before round 6 there are at most 32 pieces, already failing the first demand; round 5 also has $32/3 > 10$. Fraction comparisons can be made by multiplying both sides by 9 or 3.

29.10

Many answers: 0, 81, 27, 54, 9, 18, 63, 72. Each is an endpoint at some finite stage; after each subsequent split it is the outer endpoint of one retained third. Induction in pictures proves permanence. Do not assert that only finite-stage endpoints survive.

29.11 · optional back-pocket extension

For any finite positive piece target K and positive length budget b , both can be met by sufficiently many rounds: 2^n grows and $81(2/3)^n$ shrinks. An accessible example “more than 20 pieces, less than 12 length” first succeeds at 5. Targets with budget 0 or negative cannot succeed at a finite round, because total length remains strictly positive. Accept a chosen numeric example without requiring the general infinite-growth theorem.

29.12

At their first different letter, two addresses enter disjoint left/right thirds of a common positive-length parent, with a positive middle gap. Their final points cannot be equal. Existence and uniqueness for one infinite address use the supplied nested-interval fact. This avoids the ambiguity of ordinary terminating positional expansions.

More pieces. Less path. - solutions

29.13

The visible diagonal is LRLLLR, so the opposite letters are RLRRRL. There are many completions after those six letters. Do not claim this six-letter calculation alone defeats row 7 or any later row.

29.14

Given any proposed infinite list of infinite addresses, choose new letter n opposite to row n 's letter n . The new address differs from every row at its own numbered position; it is still a valid infinite L/R sequence and therefore names a survivor. If a point list was supplied, use each point's unique address. Hence no countable list exhausts all survivors.

29.15

Length: at round n , all never-erased points lie within 2^n closed intervals whose lengths total $81(2/3)^n$. Every two rounds multiply by $4/9 < 1/2$, so a sufficiently large even number of rounds puts that total below any positive budget. Formal outer-measure definitions using open intervals can enlarge the finitely many intervals by an arbitrarily small total allowance. List: the independent diagonal construction defeats any proposed enumeration. Small total covering length does not mean empty, finite or listable.

Offer hints only after exploration

29.2: An endpoint stays an endpoint of one surviving piece at every later round.

29.7: At each step, how many choices does an address have? Look at the first letter where two addresses differ.

29.9: Every round doubles the pieces, but each is a third as long. Keep those two changes separate.

29.14: Choose the n th new letter by inspecting the n th letter of row n .

29.15: Two rounds multiply total length by $4/9$, which is less than $1/2$.

Sources and checked scope

Sheldon Axler, Measure, Integration & Real Analysis. §2D, construction 2.74 and 2.75; results 2.76-2.79, printed 56-59 (PDF 70-73). Adult limiting construction and distinction between measure and cardinality. The 81-unit finite tasks, exact addresses and budget instance were independently constructed and checked.

<https://measure.axler.net/MIRA.pdf>

Exact checks: plans/atlas/worksheet-trial/geometry-checks.py and geometry-checks-results.json. No finite diagram exhibits zero retained length; every finite stage has positive length. No classroom pilot yet.

The addition machine has a secret

Facilitator key • independently checked mathematics; not classroom-piloted.

Assessment and satisfying core

Conditional strong for learners who enjoy logical necessity and symbolic invention. The rational-only portion is too slight to match Week 1 by itself; the two-dial construction supplies the substantial surprise. This is not an elementary manipulative worksheet.

Invent two additive machines on the same specified irrational domain that agree at every rational input yet disagree at $\sqrt{2}$, and explain why.

Prerequisites and setup

Signed rational arithmetic, use of letters for arbitrary numbers, distinction between rational and irrational numbers. The fact that $\sqrt{2}$ is irrational may be supplied.

For the continuity question: limits of rational approximations and the meaning of continuity; derivatives unnecessary.

A correct rational theorem is worthwhile, but the whole sheet's main payoff is the irrational-domain construction. Do not market early fraction calculations as a satisfactory account of the real Cauchy equation.

Materials: Pencil, two colors, optional blank input/output cards. No calculator until the final approximation challenge. Preparation: about 5 minutes.

0-8 play with chosen input claims; 8-23 two-route certificates; 23-35 the rational theorem; 35-40 break; 40-60 invent and compare irrational-domain machines. The continuity challenge can take a second meeting.

Launching and choosing an endpoint

Your partner can claim any output. You can reject it only by making two different addition routes reach the same input with different outputs.

Page 1: disprove a deceptive claim with two routes.

Page 2: prove the entire rational theorem.

Full core: construct and verify two disagreeing machines on the same domain.

What to watch for

Calling a plausible output 'forced' from a finite pattern.

Treating the irrational input as a second formal symbol without justifying unique real-number representation.

Saying the close decimal pair alone proves discontinuity.

Claiming the construction covers all real numbers.

The addition machine has a secret

Student page 1: Catch a claim with two routes

The machine accepts every rational input. Its promise is $f(x+y)=f(x)+f(y)$, and $f(1)=3$. An output is forced only if you can prove it from those promises.

G1 - prompt and solution

A technician writes these claims: $f(0)=1$; $f(-1)=-3$; $f(2/3)=21/10$; $f(5/6)=5/2$. Mark each IMPOSSIBLE or FORCED. For each decision, draw two addition routes to the same input; label the corresponding output sums.

$f(0)=1$ is impossible: additivity gives $f(0)=2f(0)$, so $f(0)=0$. $f(-1)=-3$ is forced because $f(1)+f(-1)=f(0)=0$. $f(2/3)=21/10$ is impossible: three copies would give $63/10$, whereas the input sums to 2 and $f(2)=6$; the forced value is 2. $f(5/6)=5/2$ is forced: six copies of the input sum to 5, whose output is 15, so the output is $15/6=5/2$. A claim being forced is not a proof that an unspecified entire device obeys the rule.

G1 - delayed hints

Try input 0 in both x and y .

For $f(2/3)$, compare three copies of $2/3$ with two copies of 1. For $f(5/6)$, compare six copies with five copies of 1.

G2 - prompt and solution

Choose a new rational input that looks awkward. Write a tempting wrong output. Trade with a partner: can they expose the contradiction without assuming a formula for f ?

Answers vary. For example input $-7/4$, false output -5 . Four copies sum to -7 and would give -20 , contradicting $f(-7)=-21$; the forced output is $-21/4$. For arbitrary proposed rational m/n the rule forces $3m/n$.

G2 - delayed hints

Choose m/n . Repeating it n times produces an integer you can already control.

Student page 2: A whole infinite deck

You need not test every fraction separately. Try to trap them all with one argument.

G3 - prompt and solution

Let m be any integer and n any positive integer. Explain what n copies of the input m/n add to. Use that to force $f(m/n)$. Include why your argument handles zero and negative m .

The zero argument above forces $f(0)=0$. Positive integer m is a sum of m ones, so $f(m)=3m$. Then $f(-m)=-f(m)$. For any integer m and positive integer n , $n f(m/n)=f(m)=3m$; division by n gives $f(m/n)=3m/n$. This proves every rational input at once.

The addition machine has a secret

G3 - delayed hints

First use $f(0)=f(0)+f(0)$, then $f(m)+f(-m)=f(0)$.

The same output repeated n times has to equal $f(m)$.

G4 - prompt and solution

A new input key $\sqrt{2}$ appears. Which step of your fraction proof no longer applies? Can you conclude that $f(\sqrt{2})=3\sqrt{2}$, or has that value only become a guess?

The proof requires n copies of the input to be an integer. No positive integer n makes $n\sqrt{2}$ an integer, since $\sqrt{2}$ is irrational. Therefore the rational proof does not establish its value. This does not itself prove a different value is possible; G6 supplies that construction on the enlarged specified domain.

G4 - delayed hints

Every finite sum of rational numbers is rational. Is $\sqrt{2}=m/n$ for integers m,n ?

G5 - prompt and solution

Now allow exactly the inputs $a+b\sqrt{2}$, where a and b are rational. For this task you may use the fact that each input has exactly one such pair (a,b) . Build an addition rule for these input pairs. Test your rule on $(1+2\sqrt{2})+(-3+\sqrt{2}/2)$.

Represent $a+b\sqrt{2}$ by (a,b) . Addition is $(a,b)+(c,d)=(a+c,b+d)$. The example gives $-2+(5/2)\sqrt{2}$. The set is closed under addition, negatives and rational scaling.

G5 - delayed hints

Collect the two rational coefficients separately.

Student page 3: Invent two machines that disagree

Both machines must accept exactly the inputs $a+b\sqrt{2}$, obey the addition promise, and send 1 to 3. You may design their insides.

G6 - prompt and solution

Build Machine A with $f(\sqrt{2})=0$ and Machine B with $f(\sqrt{2})=5$. Give formulas in a and b . Verify the addition promise for two arbitrary input pairs-not just for your examples.

Machine A is $f_A(a+b\sqrt{2})=3a$; Machine B is $f_B(a+b\sqrt{2})=3a+5b$. More generally c may be any real number. For pairs $(a,b),(d,e)$, $f_c((a+d)+(b+e)\sqrt{2})=3(a+d)+c(b+e)=(3a+cb)+(3d+ce)$. Thus additivity holds for arbitrary inputs, and $f_c(1)=3$. Uniqueness of the representation makes each rule a well-defined function. The fact can be proved: if $a+b\sqrt{2}=d+e\sqrt{2}$ and $b\neq e$, then $\sqrt{2}=(d-a)/(b-e)$ is rational, a contradiction; hence $b=e$ and $a=d$.

The addition machine has a secret

G6 - delayed hints

For an input $a+b\sqrt{2}$, repeated addition already determines the effect of rational multiples of 1 and of $\sqrt{2}$.

Try $f(a+b\sqrt{2})=3a+cb$. What should c be?

G7 - prompt and solution

Find one input on which the machines disagree. Then prove they agree on EVERY rational input. Explain why the uniqueness of (a,b) matters to the construction.

At $\sqrt{2}$ the outputs are 0 and 5. At any rational r its unique pair is $(r,0)$, so both outputs equal $3r$. The two machines agree on every rational, not merely on a finite tested set. Uniqueness prevents one accepted input from receiving conflicting values through alternative pair representations.

G7 - delayed hints

A rational input has pair $(a,0)$.

If the same real input had two dial settings that gave different outputs, would this be a function?

G8 - prompt and solution

Optional continuity challenge. For Machine B compare the outputs at 1.4142 and $\sqrt{2}$. You may use $1.4142 < \sqrt{2} < 1.4143$. Does that one close pair alone prove discontinuity? What happens if rational inputs approach $\sqrt{2}$ with arbitrarily small error? Which extra promise would rule Machine B out?

$f_B(1.4142)=4.2426$ and $f_B(\sqrt{2})=5$, an output gap 0.7574 for an input gap less than 0.0001. One finite close pair does not alone prove discontinuity: a continuous function may change rapidly. For any rational sequence $q_n \rightarrow \sqrt{2}$, $f_B(q_n)=3q_n \rightarrow 3\sqrt{2}$, which differs from 5. Thus Machine B is not continuous at $\sqrt{2}$ in its stated domain. Requiring continuity there would force $f(\sqrt{2})=3\sqrt{2}$; requiring continuity at all inputs would rule out B. Neither the construction nor this argument claims to define a nonstandard function on all of $\mathbb{R}$.

G8 - delayed hints

On rational inputs Machine B still gives three times the input.

Continuity concerns every desired output tolerance and sufficiently small input errors; one measured close pair is not enough by itself.

Further problems with solutions

Classify every additive function on $\mathbb{D}$ with $f(1)=3$. Let $c=f(\sqrt{2})$. Additivity forces rational homogeneity: $n f((m/n)x) = m f(x)$. Hence $f(a+b\sqrt{2})=3a+cb$. Conversely G6 verifies every such choice.

Can a continuous choice still make the two dials disagree with ordinary multiplication by 3? No. Rational approximations to $\sqrt{2}$ force $c=3\sqrt{2}$; the formula then equals $3(a+b\sqrt{2})$.

The addition machine has a secret

Sources and scope

Daniel Reem, Remarks on the Cauchy functional equation and variations of it. Introduction and §2 for additivity and regularity; the worksheet's rational proof and restricted-domain constructions are independently supplied. Source identity and abstract rechecked 2026-09-25; prior atlas review inspected the relevant discussion. <https://arxiv.org/abs/1002.3721>

Prior-use and design distinction

Original atlas rational proof upgraded to an adversarial certificate game plus a genuine family of alternative additive maps. No repeated computation table.

Exact checks: `plans/atlas/worksheet-trial/systems-checks.py`. The universal arguments above, not finite examples alone, support the general claims.

The parking robot remembers yesterday

Facilitator key • independently checked mathematics; not classroom-piloted.

Assessment and satisfying core

Strongest of these three at the Week 1 standard. A playable rule produces a false finish, demands a richer state, and leads to an all-start theorem and a geometric rotation. The optional fraction page changes the control rule instead of merely making the numbers larger.

Explain why a robot at zero can move away again, identify the ordered pair as the state, and prove the chosen pair-state cycle. The all-start theorem is the algebra continuation.

Prerequisites and setup

Track two quantities and an update order; a repeated complete state determines the future.

Letters as arbitrary numbers for the all-start proof; fractions for the half-correction experiment; distributive algebra for its global decay proof. Triangular-grid geometry is optional.

The game is an exact discrete delayed-control model. It is not a numerical approximation of a differential equation.

Materials: One token; a number line from -6 to 6 ; two erasable cards or scraps labeled OLD REPORT and CURRENT POSITION; pencils; optional counters or a body-sized number line. No cutting required.
Preparation: about 5 minutes.

0-10 free moves and familiarization with the two cards; 10-25 delayed game and false finishes; 25-40 pair-state discovery; 40-45 role swap/movement; 45-60 one proof route. Page 4 is a later or fast-group extension.

Launching and choosing an endpoint

Park the robot at zero. Each tick the controller sees yesterday's report, not today's position, and commands the opposite of that reported number. The robot adds the command to its current position.

Pages 1-2: correctly explain a false finish and a cycle in complete states.

Page 3: universal six-cycle proof or a correctly bounded geometric observation.

Page 4: prove every initial pair converges under the changed gain.

What to watch for

Overwriting CURRENT before storing it as the next old report.

A cycle claim based only on a repeated current value.

Saying all cycles have length six without exempting the zero pair.

Treating a finite half-gain experiment as proof of convergence.

The parking robot remembers yesterday

Calling the plotted standard Cartesian pair hexagon regular; the regular-hexagon claim uses the stated triangular embedding.

Student page 1: It reached zero. Is it parked?

Start OLD REPORT=1 and CURRENT POSITION=1. At every tick: (1) command = negative of OLD REPORT; (2) new position = CURRENT POSITION + command; (3) replace OLD REPORT by the position BEFORE the move, then replace CURRENT POSITION by the new position. Update the two cards together.

D1 - prompt and solution

Play until you think the story repeats. Record enough ticks to convince a partner. Circle every visit to zero. Does visiting zero mean the robot stays there?

From current $x_0=1$ with old $x_{-1}=1$, subsequent current positions are $0, -1, -1, 0, 1, 1, 0, \dots$. The visits to zero are not permanent because the old report is nonzero there. The complete pair returns to $(1,1)$ after six ticks.

D1 - delayed hints

Keep going after the first zero. What number is still on the old-report card?

D2 - prompt and solution

Change one rule: use the CURRENT position to make the command. Start at 1 again. Explain exactly what the delayed report changed.

Current feedback makes $x_{\text{new}} = x_{\text{current}} - x_{\text{current}} = 0$ in one tick and repeats zero forever. Delayed feedback uses an earlier value, so it can issue a nonzero command even at the target. This is a change in the feedback information, not a numerical approximation error.

D2 - delayed hints

With a current report, the new position is current-current.

D3 - prompt and solution

With the delayed rule restored, choose an old report and a current position that make the NEXT position zero. Can you make it remain zero forever? Find every starting pair that is already truly parked.

The next position is zero exactly when $a=b$. After that the pair is $(b,0)$, whose next current is $-b$. Hence it stays zero forever only for $a=b=0$. More generally no nonzero pair reaches $(0,0)$, because the update is invertible (D6).

D3 - delayed hints

To make the next position zero you need old=current.

The parking robot remembers yesterday

After that move, which old report will still be waiting?

Student page 2: What counts as the state?

Two robots show the same current position. Must they have the same future?

D4 - prompt and solution

Robot A has $(\text{old}, \text{current}) = (1, 1)$. Robot B has $(\text{old}, \text{current}) = (-1, 1)$. Predict one tick for each. Find your own pair of examples showing why one current reading is not enough.

A produces pair $(1, 0)$, so its next current is 0. B produces pair $(1, 2)$, so its next current is 2. Arbitrarily many other examples work: hold current fixed and choose different old values; $\text{new} = \text{current} - \text{old}$ changes.

D4 - delayed hints

Apply the two commands -1 and $+1$ to the same current position.

D5 - prompt and solution

Track the ordered pair $(\text{old}, \text{current})$, starting from $(1, 1)$. Draw an arrow for each tick. Stop only when the entire pair repeats. Why does that repeat prove the future repeats?

The ordered pairs are $(1, 1) \rightarrow (1, 0) \rightarrow (0, -1) \rightarrow (-1, -1) \rightarrow (-1, 0) \rightarrow (0, 1) \rightarrow (1, 1)$. The update is a deterministic function of the complete pair; induction then gives the same pair at every corresponding future tick. A single repeated coordinate does not support that induction.

D5 - delayed hints

The full pair is enough information to perform the next tick. Apply exactly the same rule after an identical pair.

D6 - prompt and solution

Suppose the pair after a tick is (c, d) . Recover the pair before that tick. Could two different previous pairs ever merge into the same pair?

The predecessor of (c, d) is $(c-d, c)$, since $T(c-d, c) = (c, d)$. Therefore each state has exactly one predecessor, and different pair trajectories cannot merge. In particular a nonzero state never becomes $(0, 0)$.

D6 - delayed hints

The previous current must have been c . The equation for the new current is $d = c - \text{old}$.

Student page 3: Can every start come home?

Experiments suggest a return rule. Does it work for every starting pair, even enormous or fractional ones?

The parking robot remembers yesterday

D7 - prompt and solution

Start from (a,b) . Apply $(old,current) \rightarrow (current,current-old)$. Write the successive pairs until the original pair returns. Explain why the calculation covers every real a,b .

The states $T^j(a,b)$, $j=0,\dots,6$, are (a,b) ; $(b,b-a)$; $(b-a,-a)$; $(-a,-b)$; $(-b,a-b)$; $(a-b,a)$; (a,b) . Only addition/subtraction identities were used, so this holds for arbitrary real a,b .

D7 - delayed hints

Do not choose numbers. After one tick the pair is $(b,b-a)$.

D8 - prompt and solution

Does every nonzero starting pair first return on the same tick? Check whether a return in 1, 2, or 3 ticks is possible. Why are those the only shorter cycle lengths you need to consider?

A one-tick return satisfies $a=b$ and $b=b-a$, so $a=b=0$. A two-tick return satisfies $b-a=a$ and $-a=b$; hence $-2a=a$, $a=0$ and $b=0$. A three-tick return requires $(-a,-b)=(a,b)$, again only zero. Any first return period divides six: write $6=kp+r$ with $0 \leq r < p$; because repeated p -step returns give the initial state, the six-step return would force a return at r , so $r=0$. Thus every nonzero real pair has exact period six, and zero has period one.

D8 - delayed hints

After three ticks, compare your result with (a,b) .

If T has period six, a first return length must divide six. For a proposed two-tick return solve both coordinate equations.

D9 - prompt and solution

Optional picture proof. On the triangular grid, one unit of OLD points right; one unit of CURRENT points up-left. Plot the orbit of $(1,1)$, then $(1,2)$. What rigid motion does one tick appear to make? Explain how the two unit directions could prove that motion for all pairs.

Embed (a,b) at $(a-b/2, \sqrt{3} b/2)$. T rotates this point clockwise through 60 degrees: on basis $(1,0)$ the image is $(0,-1)$, whose Euclidean coordinate is $(1/2, -\sqrt{3}/2)$; on $(0,1)$ the image is $(1,1)$, whose coordinate is $(1/2, \sqrt{3}/2)$. These are the same 60-degree rotation of the two independent basis vectors, and linearity extends the result. Therefore every nonzero orbit is a regular hexagon about the origin. Squared radius is a^2-ab+b^2 . The $(1,2)$ orbit is $(1,2), (2,1), (1,-1), (-1,-2), (-2,-1), (-1,1)$. Students without vectors can verify the two displayed orbits without claiming a full geometric proof.

D9 - delayed hints

The rightward OLD unit becomes the negative CURRENT unit. The CURRENT unit becomes their sum.

The parking robot remembers yesterday

Student page 4: A gentler correction

Optional fraction-and-algebra investigation. The report is still one tick old, but the command is now HALF of its opposite. The new pair is $(\text{current}, \text{current} - \text{old}/2)$.

D10 - prompt and solution

Start from $(1, 1)$. Predict whether a weaker correction will help or hurt. Run four ticks, then compare the whole pair with the starting pair. Run four more only if you need to test your conjecture.

Half-gain states are $(1, 1), (1, 1/2), (1/2, 0), (0, -1/4), (-1/4, -1/4), (-1/4, -1/8), (-1/8, 0), (0, 1/16), (1/16, 1/16)$. The entire pair four ticks later is $-1/4$ times the earlier pair, suggesting shrinking rather than a nonzero cycle.

D10 - delayed hints

Compare complete pairs four ticks apart, including their signs.

D11 - prompt and solution

Start instead at arbitrary (a, b) . Work out four ticks. Use what you find to prove whether EVERY starting pair approaches $(0, 0)$. A small simulation alone does not settle this.

$T_{\text{half}}^0(a, b) = (a, b)$; $T_{\text{half}}^1 = (b, b - a/2)$; $T_{\text{half}}^2 = (b - a/2, (b - a)/2)$; $T_{\text{half}}^3 = ((b - a)/2, -a/4)$; $T_{\text{half}}^4 = (-a/4, -b/4)$. Therefore $T_{\text{half}}^{(4k+r)}(a, b) = (-1/4)^k T_{\text{half}}^r(a, b)$, $r = 0, 1, 2, 3$. Each of those four fixed pairs is multiplied by a factor tending to zero, so every pair approaches $(0, 0)$. A four-tick Euclidean norm contraction alone is sufficient; one-tick contraction in every norm was not claimed.

D11 - delayed hints

After two ticks the pair is $(b - a/2, (b - a)/2)$.

Compare the fourth pair with the initial pair. Then examine the four residue classes of tick numbers separately.

D12 - prompt and solution

Design a target test: begin at $(1, 1)$ and promise that after some tick BOTH cards will have magnitude at most $1/100$, and will remain within that bound afterward. Give one certified tick, not necessarily the earliest. Explain why a small current reading alone would be inadequate.

For the chosen start, every coordinate of the states at ticks $r = 0, 1, 2, 3$ has magnitude ≤ 1 . Thus at any tick $4k + r$ both coordinates have magnitude $\leq 4^{-k}$. Taking $k = 4$ gives $\leq 1/256 < 1/100$, so tick 16 is a valid certificate for all subsequent ticks. It is not the earliest, and finding the earliest is unnecessary. A current zero can coexist with a nonzero old report that moves the next position.

The parking robot remembers yesterday

D12 - delayed hints

At ticks 0,1,2,3 for this start the largest absolute card value is at most 1.

Use the four-tick scale factor repeatedly. How many factors of $1/4$ make 1 smaller than $1/100$?

Further problems with solutions

Find a quantity preserved by full correction, using the picture. $a^2 - ab + b^2$ is preserved:

$b^2 - b(b-a) + (b-a)^2 = a^2 - ab + b^2$. Under four half-gain steps it scales by $1/16$ because each coordinate scales by $-1/4$.

For general gain k , explore what changes without claiming a theorem from plots. Characteristic roots satisfy $\lambda^2 - \lambda + k = 0$. Both have magnitude below one exactly for $0 < k < 1$. For $0 < k \leq 1/4$ they are real between zero and one; for $1/4 < k < 1$ they are conjugate with modulus $\sqrt{k}$. $k=0$ has root 1, $k=1$ has unit-modulus roots, $k < 0$ has a root > 1 , and $k > 1$ has modulus $\sqrt{k} > 1$. This extension needs quadratic roots/complex numbers and is facilitator-only unless those prerequisites hold.

Sources and scope

Åström and Murray, Feedback Systems. Chapter 7 summary, state-feedback law and closed-loop dynamics. The delayed discrete model and the worksheet's finite proofs are independently constructed. Chapter summary re-read 2026-09-25; do not attribute the exact six-cycle puzzle to this source. https://www.fbswiki.org/wiki/index.php/State_Feedback

Prior-use and design distinction

The memory pair, invertible state update, and all-start six-cycle turn a short recurrence into a substantial state-space investigation. A state graph alone would copy Week 1's representation; here the need to enlarge the state is the new idea.

Exact checks: [plans/atlas/worksheet-trial/systems-checks.py](#). The universal arguments above, not finite examples alone, support the general claims.

The route everyone wants

Facilitator notes • exact worksheet answers • not classroom-piloted

Strong worksheet candidate after replacing the original tie-based toy by a sharper instance. The six-state tie variation is a genuine dynamics question, not just another total-cost table.

Prerequisites and honest stopping points

Entry: Count to three, add costs up to 15, move one named counter at a time. An adult can read the rules.

Satisfying core: Compare a traveler's current cost with that same traveler's cost after joining the other route; distinguish personal improvement from total improvement.

Extension: Trace a directed state graph; no algebra or calculus.

Honest limit: A drawing of four occupancies suffices because travelers have identical cost rules. Different personal rules would require remembering who is where.

Materials: Three differently colored counters (R, G, B); the route mat on page 1; pencil. Optional cost counters. No scissors required.

Preparation: about 5 minutes. 0-8 move counters freely and practice charging costs; 8-22 search for stable arrangements; 22-35 explain the inefficient stable state; 35-40 stand in the two routes; 40-55 choose the tie variation; 55-60 explain one distinction. Pages are choices, not a completion quota.

Stop: Stop after 23.5 with a checked stable arrangement, a cheaper group arrangement and an explanation of the conflict. The six-state loop is an extension, not an obligatory second lesson.

Staged hints

1. For 23.2-23.3: Cover everyone except the traveler about to move. What does that traveler pay after the move, including their own arrival?
2. For 23.5: Forget the names briefly. How many travelers could QUICK have?
3. For 23.8: Compare R with RG. Which one person moved? What did that person pay before and after?

Checked solutions to every prompt

23.1

Any three arrangements are legitimate experiments. By QUICK occupancy $n=0, 1, 2, 3$, group totals are 12, 9, 10, 15. The planner chooses exactly one on QUICK; the named traveler is arbitrary.

23.2

For $n=0$ all pay 4 and a mover can get 1; $n=1$ QUICK pays 1, STEADY pays 4, and a STEADY traveler can join QUICK for 3; $n=2$ QUICK pays 3, STEADY pays 4, and neither type improves by switching ($3 \rightarrow 4$ or $4 \rightarrow 5$); $n=3$ all pay 5 and any one can move to STEADY for 4. Only $n=2$ is stable. Emphasize that joining QUICK changes its cost.

The route everyone wants

23.3

Arrows $0 \rightarrow 1$, $1 \rightarrow 2$, $3 \rightarrow 2$. No other strict-improvement arrows. Exactly one stable crowd size, $n=2$; three named stable arrangements. Compressing names is valid here because the rules treat all travelers identically. Every legal strict sequence stops after at most two moves.

23.4

Before: $R=1$, $G=4$, $B=4$, total 9. After: $R=3$, $G=3$, $B=4$, total 10. G improves by 1, R worsens by 2, B is unchanged. The group loses 1 overall. G imposes a crowding cost on R that G does not pay.

23.5

Enumerate all four occupancies: 12, 9, 10, 15. Exactly one on QUICK is the minimum 9. This covers every named arrangement because names do not affect costs. It is unstable: either STEADY traveler can lower their own cost $4 \rightarrow 3$ by joining QUICK, raising the total $9 \rightarrow 10$.

23.6

At $n=2$, one QUICK traveler may move to STEADY. That traveler's cost rises $3 \rightarrow 4$, the remaining QUICK traveler improves $3 \rightarrow 1$, and the other STEADY traveler stays 4. Total $10 \rightarrow 9$. Individual stability guarantees only the absence of a personally profitable one-person move; it does not guarantee the least group total.

23.7

With STEADY 3: $n=0$ has a strict move $3 \rightarrow 1$; $n=1$ has no strict move because joining QUICK costs 3, a tie; $n=2$ has no strict move because leaving QUICK costs 3, a tie; $n=3$ has a strict move $5 \rightarrow 3$. Thus $n=1$ and 2 are stable under strict moves. With ties allowed, a traveler may alternate between QUICK and STEADY while one other remains on QUICK, producing endless play. No simultaneous moves are needed.

23.8

The only edges are $R \leftrightarrow RG \leftrightarrow G \leftrightarrow GB \leftrightarrow B \leftrightarrow RB \leftrightarrow R$. A move adds or removes one name, switching between one and two QUICK travelers. The mover pays 3 on either road. Opposite pairs such as R and GB change three names and are illegal. The printed cyclic order offers one valid six-state tour; students may traverse it either way or begin anywhere.

23.9

Singleton boxes total $1+3+3=7$; two-name boxes total $3+3+3=9$. The six-state loop alternates 7, 9, 7, 9, 7, 9. The moving traveler stays at 3; the other QUICK traveler changes $1 \leftrightarrow 3$. The group need not improve when the mover alone does not worsen.

The route everyone wants

What this really teaches

An exact finite atomic congestion game. The base is stronger than the atlas instance because the inefficient equilibrium has no indifferent escape: the efficient pattern is strictly unstable. In the base, a congestion potential has values 12, 9, 8, 9 across crowds 0, 1, 2, 3 and decreases at each strict personal improvement; total cost has different values. The worksheet does not need this theorem or name.

Prior use: The current route worksheets study edge traversal. Week 1 studies undirected tiling flips. This asks who bears a move's cost and contrasts strict versus weak improvement. The six-state shape is not claimed novel by itself.

Source adaptation: Adapted mathematical framework from the AP-23 source below; all numerical games, graph prompts and solutions here are new independently checked instances. No claim of a classroom pilot.

Giacomo Bonanno, Game Theory. Section 1.6: Nash equilibrium; finite congestion instance checked directly. <https://arxiv.org/pdf/1512.06808> Evidence status: named source scope inspected; displayed finite result proved directly here

Verification: `plans/atlas/worksheet-trial/decisions-checks.py`; exact enumeration and rational calculations in `decisions-checks-results.json`. Full battery state tables are in that checked data; they are deliberately absent from the student work.

The robot's five-day battery

Facilitator notes • exact worksheet answers • not classroom-piloted

Strong for children who enjoy planning. The original three-day counterexample was too slight. Five offers, converging histories, a compact backward certificate and a changed ending produce real substance without a 30-cell arithmetic drill.

Prerequisites and honest stopping points

Entry: Count and spend up to 5 battery tokens; add rewards up to 18. Read a five-day order, with oral help if desired.

Satisfying core: Keep a fixed resource budget and justify an upper bound by cases.

Extension: Reason backward about best possible future scores; no algebra or calculus.

Honest limit: All offers are visible in advance. This is not a strategy for an unknown future. The future depends on day and battery, while total score also includes points already earned.

Materials: Five battery counters, about 20 reward counters or a written score, and five printed offer cards. Cross off each day as it passes. Scissors optional: cards can remain on the sheet.

Preparation: about 7 minutes. 0-10 act out a couple of plans; 10-25 improve them; 25-35 compare the two histories; 35-40 swap planner/robot roles; 40-55 use the backward certificate or changed ending; 55-60 explain why the bound covers all choices.

Stop: A feasible 18-point plan plus a convincing take/skip upper bound is a complete investigation. The changed ending can be a separate revisit. Do not turn the six-state certificate into a mandatory full 30-cell table.

Staged hints

1. For 21.1-21.3: Try a plan that skips day 1. A maximum needs both a plan and a reason no better plan can exist.
2. For 21.4: Put earned points in a separate pile. Which objects can affect what jobs the robot is still able to take?
3. For 21.7-21.8: Every plan either takes today's job or skips it. If it takes the job, what exact day and battery will it have next?

Checked solutions to every prompt

21.1

Many feasible plans. Unique optimum is days 2, 4, 5: battery $2+1+2=5$, points $6+4+8=18$. Let students find it before offering structural hints.

The robot's five-day battery

21.2

Chronological take-when-possible chooses days 1, 2, 4, spends 5, earns $5+6+4=15$. Highest reward first chooses day 3 (9 points, cost 3), then day 5 (8, cost 2), earning 17. Days 2, 4, 5 earn 18. The second rule here chooses among all visible offers, then schedules selected offers on their proper days. Neither counterexample says every greedy rule always fails.

21.3

Accept an explicitly feasible candidate and an honest distinction between finding a plan and excluding better plans. Possible future proof ideas: list all feasible subsets, split into take/skip cases, or develop reusable best-future cards. A guessed maximum is not yet a certificate.

21.4

X: battery 3, earned 5. Y: battery 3, earned 6. Both should skip day 3 and take days 4, 5 for 12 more points. Final totals 17 and 18. They share the best future decisions and extra score because day and battery coincide. They do not share the points already earned. An earlier score is an additive constant in the future optimization, so it does not change the maximizing continuation.

21.5

For battery 0, 1, 2, 3, 4, 5, best extra scores are 0, 4, 8, 12, 12, 12. With 0 take none; with 1 take day 4; with 2 take day 5; with 3+ take both. Rewards are positive and each remaining job is available at most once. More than 3 battery buys no new job; the maximum remains 12.

21.6

Take day 3: gain 9 now, no battery remains, extra total 9. Skip: retain 3 for days 4, 5, whose proved value is 12. Skip is better. Earlier earned points add equally to both alternatives, so they cannot change this comparison.

21.7

Before day 3, battery 1: cannot take cost 3, so value 4 from day 4 card. Battery 3: $\max(9+0, 12)=12$. Battery 5: $\max(9+8, 12)=17$. The 8 is day 4 value with 2 battery, achieved by skipping day 4 and taking day 5. This is why one must track battery as well as day.

21.8

(a)Day 2, battery 3: take gives $6+V(\text{day } 3, 1)=10$, skip gives $V(\text{day } 3, 3)=12$, so value 12. (b)Day 2, battery 5: take $6+V(\text{day } 3, 3)=18$, skip $V(\text{day } 3, 5)=17$, so value 18. (c)Day 1, battery 5: take $5+V(\text{day } 2, 3)=17$, skip $V(\text{day } 2, 5)=18$, so value 18. (d)Trace skip 1, take 2, skip 3, take 4, take 5. At each card the take/skip split exhausts all legal possibilities, and the later values have already been proved. Therefore 18 is both an upper bound and achieved, not just the largest tried plan.

The robot's five-day battery

21.9

There are exactly two optimal plans, days 2, 3 and days 1, 2, 4, each worth 15. With day 3 taken, remaining battery 2 can add at most 6 (day 2 beats day 1=5, day 4=4, day 5=3), giving at most 15. If day 3 skipped and day 2 taken, there are 3 battery left for days 1, 4, 5: day 1+day 4 gives 9, larger than day 4+day 5=7, so total at most 15. If both day 3 and day 2 skipped, all of days 1, 4, 5 fit at cost 5 and pay 12. These cases cover every plan. The day 4 help-card row changes to 0, 4, 4, 7, 7, 7; the affected later-to-earlier cards must be recomputed. Full changed value rows are in decisions-checks-results.json. It is legitimate to ignore the old table and give the three-case certificate instead.

What this really teaches

A deterministic finite-horizon dynamic program; the worksheet creates common subproblems before naming a backward method. $V(d, e) = \max(V(d+1, e), \text{reward}_d + V(d+1, e - \text{cost}_d))$ over legal actions. The proof is backward induction. Converging histories motivate state sufficiency and, in forward search, dominance by larger earned score.

Prior use: Weeks 7-8 use backward winning states in adversarial games. This task uses a resource state, accumulated reward and optimization against constraints, with no opponent. These are related proof habits but different questions.

Source adaptation: Source framework from AP-21. The five-job instance, all state cards and changed-ending classification were checked independently by exhaustive subset enumeration for every day/battery state.

Jeffrey Chasnov, Mathematical Biology. Chapter 7.3: dynamic programming located; this finite resource recurrence is self-contained. <https://www.math.hkust.edu.hk/~machas/mathematical-biology.pdf> Evidence status: dynamic-programming chapter located in authored contents; this finite decision recurrence has a complete independent proof

Verification: plans/atlas/worksheet-trial/decisions-checks.py; exact enumeration and rational calculations in decisions-checks-results.json. Full battery state tables are in that checked data; they are deliberately absent from the student work.

Make a maze. Then turn it inside out.

Facilitator preparation, staged hints and complete solutions

What counts as a satisfying result

One valid maze and a reason all valid mazes remove 4 roads, OR a shortest 4 swaproute with a lower bound.

Strong candidate for the Week 1 richness standard. Four pages are too much for many one-hour groups; use 1-2 plus either 3 or 4. The courtyard page needs physical tracing; do not force a general topology proof.

Prerequisites

Entry: Trace a route; count to 12; distinguish a loop from going along one road and back. Oral directions suffice.

Core explanation: Choose routes and check all rooms; explain a leaf-removal certificate or a shortest sequence of swaps. No algebra required.

Extension: Track the outside as one region; compare two planar networks. A complete theorem for arbitrary planar graphs is optional, not claimed from four examples.

Materials, preparation and pace

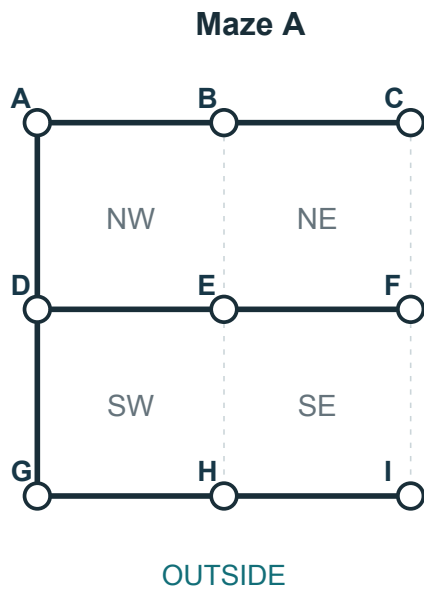
12 toothpicks or removable road strips per pair; 9 counters on the room dots; pencil and eraser; optional transparent sheet/highlighter for courtyard flood. Preparation: about 10 minutes.

0-8 free road designs; 8-13 demonstrate a loop and a disconnected room; 13-28 maze challenge and a proof; 28-33 walk a human tree; 33-50 choose swaps OR courtyard view; 50-60 share a certificate. Four pages are a menu for one or two meetings, not a required pace.

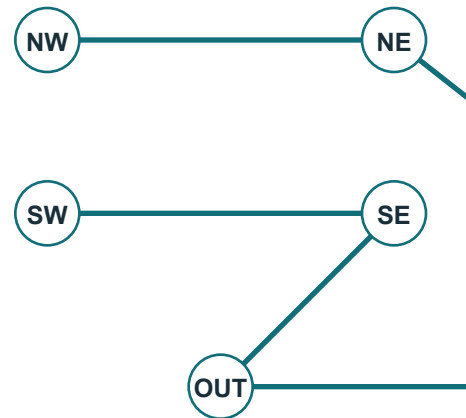
Week 10 asks Euler/postman traversal; this uses spanning trees and planar complementarity. The swap lower bound resembles Week 1 intentionally, but the maze objects and dual representation are new. Atlas square+diagonal instance has been enlarged and replaced.

Make a maze. Then turn it inside out. - solutions

Prompt numbers match the student pages; back-pocket questions are noted.



Complementary region tree



25.1

Many answers. Maze A keeps AB, BC, DE, EF, GH, HI, AD, DG, removing BE, EH, CF, FI. It connects all nine rooms and has no cycle. There are 192 successful kept edge sets in this fixed grid. Do not require students to enumerate them.

25.2

Maze B keeps AB, BC, AD, DG, BE, EH, CF, FI, removing DE, EF, GH, HI. Both remove 4 roads. Any other distinct verified tree works.

25.3

Deleting one road of a loop preserves all possible reachability: the rest of that loop gives a detour. Continue until no loops remain. Every resulting connected acyclic graph on 9 vertices has 8 edges (proved on page 2), so $12 - 8 = 4$ removals are necessary and sufficient. Before the formal count, accept a correct paired room/road certificate explained with counters.

25.4

The perimeter uses 8 roads, but it leaves E isolated and also retains the outer 8-edge loop. Having the right edge count alone is not enough.

Make a maze. Then turn it inside out. - solutions

25.5

For Maze A one valid leaf sequence is C, B, F, E, I, H, G, D, leaving A. Students may start from any current leaf; checks must use the graph after previous removals. For Maze B one sequence is G, D, H, E, I, F, C, B, leaving A.

25.6

Each leaf deletion removes exactly 1 room and 1 road, preserves connectivity among remaining rooms, and cannot create a loop. Eight room deletions leave one room with no roads, so every successful maze had 8 roads. The starting 12 roads therefore lose 4.

25.7

Take a longest simple path in the finite tree. An endpoint cannot have a neighbor off the path, since that extends it; it cannot have another neighbor already on the path, since that makes a loop. Thus the endpoint has degree 1 whenever more than one vertex remains. This is a picture proof, not a demand for the formal word “induction.”

25.8

Use only the 12 original grid edges. Do not slide rooms, invent diagonal roads, or regard rotated drawings as an allowed swap. A valid solution is supplied under 25.9.

25.9

Four swaps: restore BE/remove DE; restore CF/remove EF; restore EH/remove GH; restore FI/remove HI. Every intermediate graph is a tree; exact check script verifies it. Other shortest sequences are valid.

25.10

The target has four roads BE, EH, CF, FI that the start lacks. One swap restores at most one of them, so at least 4 swaps are necessary; the displayed plan meets that bound.

25.11

A tree has exactly one simple route between any two rooms. Adding one road creates exactly one loop: that road plus the old route between its endpoints. Remove any other edge on that loop, and the graph is a tree again. Removing an edge outside that loop disconnects the graph. If the new road is itself removed, the state is unchanged; the task excludes that empty move.

25.12

Yes for every successful maze. For Maze A the removed edges are BE, EH, CF, FI; the region links are NW-NE, SW-SE, NE-OUT, SE-OUT. All 5 region dots are connected by 4 links with no loop. Water cannot slip through a retained wall vertex: regions cross only through a removed edge.

Make a maze. Then turn it inside out. - solutions

25.13

Yes. If a collection of courtyards cannot reach OUT, its outer boundary consists of kept roads and contains a loop; a successful maze has no such loop. The picture is sufficient for this rectangular grid. The stronger dual-tree theorem also needs the primal connected condition, ensuring the removed-region links have no dual cycle.

25.14

The complementary network is another tree: 5 region dots and 4 links. For Maze B its links are NW-SW, NE-SE, SW-OUT, SE-OUT. All 4096 edge subsets were checked: primal kept roads form a tree exactly when complementary dual region links form a tree. Parallel outside links are distinct roads, so do not merge them in a broader example.

25.15 · optional back-pocket extension

For an m -by- n array of square courtyards there are $(m+1)(n+1)$ rooms and $m(n+1)+n(m+1)$ roads. The required number removed is mn , matching one less than the $mn+1$ regions including outside. A 3 by 2 courtyard example has 12 rooms, 17 roads, needs 6 removals. Keep all horizontal rows and one vertical boundary to construct a maze. Its complement is a dual tree. The general statement is planar graph duality, not a claim about arbitrary crossing networks.

Offer hints only after exploration

25.3: Can a road on a loop be removed without blocking any room?

25.6: Pair each disappearing road with one disappearing room. Which room has no partner?

25.10: Count roads that Maze B needs and Maze A is missing.

25.13: If some courtyards stay dry, trace the outside boundary of that dry patch.

Sources and checked scope

Allen Hatcher, Algebraic Topology. Chapter 0; §1.A, Proposition 1A.2, printed 84-85 (PDF 92-93). Adult graph/fundamental-group connection. This specific 3×3 maze, swaps and dual worksheet are independently designed/verified.

<https://pi.math.cornell.edu/~hatcher/AT/AT.pdf>

Exact checks: plans/atlas/worksheet-trial/geometry-checks.py and geometry-checks-results.json. No classroom pilot yet; no claim that every random atlas card would support this expansion.

Can you trust the cooling simulator?

Facilitator key • independently checked mathematics; not classroom-piloted.

Assessment and satisfying core

Strong calculus investigation after redesign; weak as an elementary recurrence-only worksheet. The payoff is designing the numerical method under a cost and accuracy constraint, with a proved optimal step allocation. It deliberately requires more mathematics than AP-26.

Derive Euler's update from a tangent model, distinguish decaying output from accurate output, and prove the best placement of a limited number of updates.

Prerequisites and setup

Calculus to understand the differential model and the approximation; complete-the-square algebra for two-step optimization. A calculator evaluates powers and exp.

For the global six-update lower bound, use or prove the equal-factor product inequality (AM-GM). No advanced ODE theory is required.

Do not lower the main gate to signed arithmetic. A recurrence can be explored with less preparation, but identifying and improving a numerical approximation of cooling is the satisfactory core.

Materials: Pencil, ruler, two colors, scientific calculator. No hot liquid or temperature equipment. Preparation: about 5 minutes.

0-12 curve and tangent experiments; 12-25 failures and all-step-size classification; 25-40 split-step design; 40-45 pause; 45-60 cost/accuracy challenge. Page 4's lower bound and backward method can be a second session.

Launching and choosing an endpoint

The mathematical cooling model is $y' = -y$, $y(0) = 8$, where y is degrees above room temperature and time is measured in fixed chosen units. The simulator uses the slope at the beginning of each step as if that slope stayed constant for the whole step. When should we distrust it?

Page 2: prove the full qualitative classification and a precise accuracy counterexample.

Page 3: prove the unique optimal two-update schedule.

Page 4: prove six updates are necessary and sufficient under the stated 10% target.

What to watch for

Using the same number of updates as if it meant the same elapsed time.

Calling bounded oscillation 'settling'.

Asserting stable means accurate.

Can you trust the cooling simulator?

Assuming improving the final numerical value improves accuracy without showing it lies below the exact value.

Transferring scalar commuting-step formulas to general nonlinear or matrix systems.

Student page 1: A computer walks along tangents

Our model is $y' = -y$, $y(0) = 8$. Here y is the difference from room temperature, not absolute temperature. The exact model solution is $y(t) = 8e^{-t}$.

C1 - prompt and solution

Check the supplied exact solution using its derivative and initial value. What must remain true about its sign and direction of change for every finite $t \geq 0$?

$y'(t) = -8e^{-t} = -y(t)$ and $y(0) = 8$. For every finite $t \geq 0$ the solution is strictly positive and its derivative is strictly negative. It approaches zero without attaining or crossing it.

C1 - delayed hints

Differentiate $8e^{-t}$. Exponentials are positive.

C2 - prompt and solution

At a state (t, y) , the simulator follows the current tangent slope for a duration $h > 0$. Derive its next value. Starting from $(0, 8)$, draw ONE simulator step for $h = 1/2$, $3/2$, and 3 . Label each endpoint with its actual elapsed time. Which steps violate the model's qualitative behavior?

Euler gives $y_{\text{new}} = y + h(-y) = (1-h)y$. At the initial point, the tangent is $y = 8 - 8t$. The endpoints are $(1/2, 4)$, $(3/2, -4)$, and $(3, -16)$. The last two fall below the room-temperature level even though the exact solution stays above it. All three differ quantitatively from the curved exact solution.

C2 - delayed hints

A tangent with slope $-y$ changes its height by $-hy$ over horizontal distance h .

C3 - prompt and solution

A teammate says a negative simulated value means negative absolute temperature. Correct the interpretation, then explain why a below-room-temperature value is still wrong for this particular model and start.

A negative y means below room temperature; absolute temperature would add the room's absolute value. Nonetheless this cooling model with $y(0) > 0$ never goes below room temperature, as its exact positive exponential solution proves. The numerical approximation invented the crossing.

Can you trust the cooling simulator?

C3 - delayed hints

Zero y means room temperature. The exact solution for this start never crosses zero.

Student page 2: Settling down is not enough

Keep the SAME positive step size h at every update. Say that the simulator settles if its values tend to zero as the number of updates grows.

C4 - prompt and solution

Rewrite the update as multiplication by one number. Classify ALL $h > 0$ into: positive decreasing values; hits zero in one step; alternating values that settle; a persistent bounded oscillation; growing magnitude. Include every boundary value.

For $0 < h < 1$ the multiplier lies in $(0, 1)$, so the values stay strictly positive and decrease to zero. At $h = 1$ the first update becomes zero and all later values are zero. For $1 < h < 2$ the multiplier is in $(-1, 0)$, giving alternating signs with shrinking magnitudes and convergence to zero. At $h = 2$ it is -1 , giving $8, -8, 8, \dots$, bounded but not tending to zero. For $h > 2$ the multiplier is below -1 and magnitude grows. Thus settling means $0 < h < 2$; preserving strict positivity at every finite step means $0 < h < 1$. Some texts define absolute stability by boundedness, which includes the boundary $h = 2$; this sheet deliberately uses the explicit word 'settles'.

C4 - delayed hints

The multiplier is $1 - h$. Compare it with $1, 0, -1$.

Treat $h = 1$ and $h = 2$ separately.

C5 - prompt and solution

Build a counterexample to 'If the simulator settles, its answers are accurate.' Use the model value at the same elapsed time, not the same step number. Quantify the relative error $|\text{approximation} - \text{exact}| / \text{exact}$.

Take $h = 1$. The simulation settles immediately at zero. At elapsed $t = 1$ it predicts 0 whereas the model gives $8/e \approx 2.943035529$, so relative error is $1 = 100\%$. This disproves the claim without confusing long-time decay and fixed-time accuracy.

C5 - delayed hints

At $h = 1$, what does the simulator predict at $t = 1$? Does the exact model reach zero then?

Can you trust the cooling simulator?

C6 - prompt and solution

Change the physics to $y' = -2y$ with the same initial value. Without calculating a long sequence, identify the positive step sizes that make the simulator settle, and those that keep every value strictly positive.

With $y' = -2y$, Euler multiplier is $1 - 2h$. Settling requires $|1 - 2h| < 1$, equivalent to $0 < h < 1$. Strictly positive decreasing iterates require $0 < 1 - 2h < 1$, equivalent to $0 < h < 1/2$. At $h = 1/2$ the simulator reaches zero immediately. The thresholds depend on the rate constant; h alone is not dimensionless in a general physical model.

C6 - delayed hints

The multiplier becomes $1 - 2h$. Tending to zero requires its absolute value to be below 1.

Student page 3: Spend two updates wisely

Return to $y' = -y$. You must estimate the value at elapsed time $t = 1$ using exactly TWO Euler updates. You choose the first duration h , with $0 < h < 1$. The second duration is $1 - h$. Start from $y = 8$.

C7 - prompt and solution

Try two different choices of h , then derive a formula for the final simulated value in terms of h . Compare your choices at $t = 1$.

The final value is $8(1-h)(1-(1-h)) = 8h(1-h)$. Example $h = 1/4$ gives $3/2$; $h = 1/2$ gives 2. Both end at $t = 1$. Exchanging the order of the two durations leaves their multiplier product unchanged for this scalar linear equation.

C7 - delayed hints

Multiply the two step multipliers: $(1-h)(1-(1-h))$.

C8 - prompt and solution

Which choice of h gives the closest answer to $8/e$? Prove it beats EVERY other two-step choice, not just the choices you tried. Explain why maximizing the simulated final value really does minimize the error here.

$8h(1-h) = 2 - 8(h - 1/2)^2 \leq 2$, with equality only at $h = 1/2$. Since $8/e \approx 2.943 > 2$, every allowed two-step estimate lies below the exact value; the largest estimate is therefore the closest. The unique optimal schedule is two half-steps, final value 2 and relative error $1 - e/4 \approx 32.043\%$. Optimizing within a poor budget still leaves a poor approximation.

C8 - delayed hints

Rewrite $h(1-h)$ as $1/4 - (h - 1/2)^2$.

The maximum simulated value is 2. Compare that with $8/e$.

Can you trust the cooling simulator?

C9 - prompt and solution

An engineer suggests replacing one duration H by two positive durations u and v with $u+v=H$. Compare the two resulting multipliers algebraically. What improves when $0 < H \leq 1$? Which conclusion about accuracy would require comparison to the exact model?

Two steps give $(1-u)(1-v)=1-H+uv$, exceeding the single-step multiplier $1-H$ by $uv > 0$. For $0 < H \leq 1$ all factors are nonnegative. Also $1-s \leq e^{-s}$ for $s \geq 0$, so $(1-u)(1-v) \leq e^{-H}$; the split-step value lies between the one-step estimate and the exact multiplier for positive starting y . Thus splitting improves the local final-time approximation in this scalar case. The factor comparison alone only proves the numerical value increased; the exact-solution bound is needed to conclude accuracy improved. It is not a theorem for arbitrary ODEs.

C9 - delayed hints

Expand $(1-u)(1-v)$, and subtract $1-H$.

For $s \geq 0$, $1-s \leq e^{-s}$; derive it from a tangent or a derivative if needed.

Student page 4: An accuracy budget

You may choose any positive step durations adding to 1. Goal: approximate $y(1)=8/e$ with relative error at most 10%, using as few Euler updates as possible. A calculator is allowed.

C10 - prompt and solution

Find a schedule that meets the target. Equal steps are a sensible experiment, but unequal steps are allowed. State your durations, final estimate, and relative error.

Six equal durations $h_i=1/6$ give $Y=8(5/6)^6=15625/5832 \approx 2.679183813$. Relative error is $1-e(5/6)^6 \approx 0.089652916$, below 10%. Five equal steps give $8(4/5)^5=2.62144$ and relative error ≈ 0.109273410 , above 10%. C11 excludes all other schedules with at most five.

C10 - delayed hints

For n equal durations, the estimate is $8(1-1/n)^n$. Compare it to $8/e$.

Can you trust the cooling simulator?

C11 - prompt and solution

Prove no schedule with fewer updates than your proposed minimum can meet the target. You may use this fact: for a fixed sum, a product of nonnegative numbers is largest when all factors are equal. Identify the factors and their sum. Explain how schedules with still fewer updates fit your argument.

For a schedule of $m \leq 5$ positive durations summing to 1, add $5-m$ zero-duration placeholders. The estimate remains $8 \prod (1-h_i)$. The five factors are nonnegative and sum to $5-1=4$, so their product is at most $(4/5)^5$. Hence every such schedule gives $Y \leq 2.62144$, while 90% of the exact value is $7.2/e \approx 2.648731976$. All estimates are below the exact value by the same $1-h \leq e^{-h}$ bound, and the upper bound already misses the allowed interval. Therefore at least six updates are necessary, and C10 attains six. The supplied product fact can be proved by a maximum argument on the compact nonnegative fixed-sum simplex: if a maximizing positive product had unequal factors u, v , replacing both by $(u+v)/2$ preserves the sum and increases their product by $(u-v)^2/4$, a contradiction; the all-equal candidate has positive product, so a maximizer cannot have a zero factor.

C11 - delayed hints

If there are fewer than five updates, insert zero-duration placeholders, which multiply by 1 and change nothing.

With five factors $(1-h_i)$, the sum is 4. The largest product is $(4/5)^5$.

C12 - prompt and solution

Optional repair. Another simulator chooses the slope at the END of each step: $y_{\text{new}} = y_{\text{old}} - h \cdot y_{\text{new}}$. Solve for y_{new} . Does it remain positive and settle for every $h > 0$? Does that guarantee 10% accuracy with one step to $t=1$?

Backward Euler gives $y_{\text{new}} = y_{\text{old}} / (1+h)$, with multiplier in $(0,1)$ for every $h > 0$. Thus every positive start remains positive and settles for any fixed positive h . At one step $h=1$ it predicts 4, while exact is $8/e \approx 2.943$; relative error is $e/2 - 1 \approx 35.914\%$, still above 10%. Removing the qualitative failure does not remove the accuracy requirement.

C12 - delayed hints

Move $h \cdot y_{\text{new}}$ to the left. The multiplier becomes a reciprocal.

Further problems with solutions

For $y' = -ay$ and total time T with $aT \leq 1$, solve the best n -step allocation. For nonnegative h_i summing to T , multipliers $1 - a h_i$ are nonnegative with sum $n - aT$. Their product is maximal when $h_i = T/n$, giving $y_0(1 - aT/n)^n$. Since each multiplier is at most $e^{-a h_i}$, the numerical final value is below the exact solution for $y_0 > 0$, so this maximizes accuracy as well. The restriction $aT \leq 1$ guarantees no negative factors for arbitrary permitted partitions; other regimes require additional constraints.

Can you trust the cooling simulator?

Prove the one-step lower estimate without already assuming a Taylor expansion. For $g(s)=e^{(-s)}-(1-s)$, $g(0)=0$ and $g'(s)=1-e^{(-s)}\geq 0$ for $s\geq 0$. Hence $g(s)\geq 0$.

Sources and scope

Driscoll and Braun, Fundamentals of Numerical Computation. §11.3, Example 11.3.1 and equations (11.3.5)-(11.3.6), Euler multiplier; Definition 11.3.1 uses boundedness, including the boundary. Relevant exposition read 2026-09-25. Two-step optimization and six-update accuracy challenge independently designed and proved. <https://fncbook.com/absstab-diffusion/>

Prior-use and design distinction

The atlas's multiplier exercise becomes a design problem: a tangent simulation with physical constraints, fixed-time comparisons, an optimal allocation of two updates, and an exact minimum update budget under an accuracy target. This is intentionally a calculus sheet.

Exact checks: `plans/atlas/worksheet-trial/systems-checks.py`. The universal arguments above, not finite examples alone, support the general claims.

A fair price for a wandering token

Facilitator notes • exact worksheet answers • not classroom-piloted

Worth making, but the honest complete version needs halves/quarters and first-step reasoning. Playing with a coin alone is only an entry. The ferry variation earns its third page by separating path duration from endpoint chance.

Prerequisites and honest stopping points

Entry: Move a counter one step and follow heads/tails. Counting trials is optional.

Satisfying core: Halves and quarters; average two quantities, or use equal height gaps and a balance argument. Distinguish a trial from an exact probability.

Extension: Multiply fractions for the optional almost-sure stopping argument; limits are an adult extension. No calculus required.

Honest limit: A few trials do not prove the probabilities. Expected reward and average-of-neighbors need explanation; do not call this a complete probability lesson for a child who is only following the moves.

Materials: A coin, a counter, pencil, and optionally stacking cubes for chance heights. Use a fair coin model: each toss has independent half-chances. A physical coin is an implementation, not a proof of fairness.

Preparation: about 5 minutes. 0-8 explore paths; 8-15 record at most 4 completed trials; 15-35 build and justify the height map; 35-40 walk the track physically; 40-55 try the ferry; 55-60 compare exact evidence and experiment.

Stop: A justified 0, 1, 2, 3, 4 price skyline and the quarter/half/three-quarter chances. A younger child who only plays has explored the object but has not completed this argument; record that distinction honestly.

Staged hints

1. For 01.4: Imagine many tokens, half arriving at one neighbor and half at the other. What is their average value?
2. For 01.5-01.6: A balanced middle height has the same rise from its left neighbor as to its right neighbor. Follow that condition across the whole track.
3. For 01.7-01.9: A shortcut replaces the two possible next prices. Check their average, not how far the token travels.

Checked solutions to every prompt

01.1

Valid records follow H:right, T:left and stop on first reaching a shore. Higher starts have larger exact winning chances: $1/4$, $1/2$, $3/4$. No interior start guarantees a win; an appropriate finite run of tails reaches 0 with positive probability. Any trial pattern, including four identical results, is acceptable experimental evidence and does not refute the exact probabilities.

A fair price for a wandering token

01.2

Predictions are provisional here. Exact fair prices later are 1, 2, 3 for starts 1, 2, 3. Avoid grading guesses by proximity. Ask whether the reason distinguishes an endpoint prize from the number of tosses.

01.3

The pairing argument is valid: reflect a path across the middle and interchange heads/tails to exchange its two endpoint outcomes while preserving probability. It does not make an individual path win. A full claim of half each also uses eventual arrival with probability 1; the elementary first-step map below independently gives the winning probability. The first statement confuses symmetry with certainty.

01.4

Fair price is $(a+b)/2$: there are two equally likely next situations, so expected future prize averages their values. Integer-price balance can be written $2 \times \text{HERE} = a+b$, avoiding fraction notation initially. If p_i is the eventual winning probability, the same conditioning gives $p_i = (p_{\text{left}} + p_{\text{right}})/2$. The existence of a well-defined eventual hitting event makes this equation legitimate, not an inference from simulation.

01.5

Heights 0, 1, 2, 3, 4 balance every interior place. No other skyline with these endpoint heights works. Changing one interior height while freezing the others breaks at least that point's balance. To exclude coordinated changes as well, use 01.6, not just this perturbation test.

01.6

If $2b = a+c$, then $b-a = c-b$: the two adjacent gaps agree. Applying this at 1, 2, 3 forces all four gaps equal. Their total is 4, so each gap is 1. Thus prices 1, 2, 3 are uniquely forced. The prize is 4 on a win and 0 otherwise, so $\text{price} = 4 \times \text{winning chance}$: chances $1/4, 1/2, 3/4$. The proof is a local-to-global uniqueness argument, not curve-fitting.

01.7

The prices remain 0, 1, 2, 3, 4. At 1: $(0+2)/2=1$; at 2: $(0+4)/2=2$; at 3: $(2+4)/2=3$. Since the ferry dock terminates immediately at a shore on its next toss, its price 2 is directly fixed. The prices at 1 and 3 then follow directly; these equations have no alternate solution. All three winning chances are unchanged.

01.8

Paths and duration changed; winning chances did not. From 2 the ferry ends after exactly one toss, whereas an ordinary walk from 2 cannot reach a shore before two tosses. Yet both give chance $1/2$ of winning. With the ferry, no game from an interior start lasts more than two tosses. Original games can be arbitrarily long, though terminate with probability 1. Faster arrival is not by itself a higher chance of the right endpoint.

A fair price for a wandering token

01.9

If a changed start is i and destinations are a, b , old prices stay balanced exactly when $a+b=2i$, with $0 \leq a, b \leq 6$ and $a, b \neq i$. Valid new choices include start $2 \rightarrow 0/4$, start $3 \rightarrow 0/6$ or $1/5$, start $4 \rightarrow 2/6$. Starts 1 and 5 have no distinct-destination longer jump within the board; their only unordered pair is $0/2$ or $4/6$. Swapping H/T labels makes the same average. A completed answer must state both destinations, not just the favorable jump. For the single modified site, eventual absorption still holds: from any site, repeatedly choosing a rightward destination reaches 6 in at most 6 moves, probability at least $1/64$ per independent block. The unchanged linear prices satisfy the first-step system uniquely. For elementary learners, validating the local balance is the intended stopping point; claiming a general stopping theorem is unnecessary.

What this really teaches

Discrete harmonic functions and a finite Dirichlet problem. The linear solution arises from equal first differences. Almost-sure absorption in the ordinary walk: a block of 3 all-heads tosses reaches 4 from any interior state, with probability $1/8$; survival after k independent blocks is at most $(7/8)^k \rightarrow 0$. Finite first-step equations already suffice to derive the probability of the eventual winning event; absorption additionally guarantees the two shore events exhaust outcomes up to probability 1. Expected toss counts, not required on student pages: ordinary $(0, 3, 4, 3, 0)$, ferry $(0, 3/2, 1, 3/2, 0)$.

Prior use: Week 1's extension samples an ongoing tiling-flip walk and asks about visit frequencies/return times. This has absorbing shores, exact hitting probabilities, an affine harmonic function and a changed transition preserving the boundary chances. Do not count mere coin-tossing as novelty.

Source adaptation: Source framework from AP-01. Ferry, price representation and shortcut design are independently checked adaptations. Prices are imaginary prize counters, not financial guidance.

Grinstead and Snell, Introduction to Probability. Chapters 4 and 12.2.

<https://math.dartmouth.edu/~prob/prob/prob.pdf> Evidence status: relevant exposition inspected; finite instance independently derived below

Verification: [plans/atlas/worksheet-trial/decisions-checks.py](#); exact enumeration and rational calculations in [decisions-checks-results.json](#). Full battery state tables are in that checked data; they are deliberately absent from the student work.

Send it with fewer bits

Facilitator notes • exact worksheet answers • not classroom-piloted

Very strong worksheet candidate. It has physical message strips, a real ambiguity, freely designed codes, a finite optimality certificate and a frequency change that reverses the winning tree.

Prerequisites and honest stopping points

Entry: Match four symbols, read 0/1 strings left to right and count bits. An adult can read instructions.

Satisfying core: Repeated addition for weighted length, branching diagrams, and explaining why an example works for every concatenation.

Extension: Classify full binary trees with four leaves and compare costs; no probability or logarithms needed.

Honest limit: Optimality is proved among binary prefix codes, not all conceivable communication schemes. No free spaces, timing gaps, punctuation or hidden lengths are allowed.

Materials: Four distinct simple icons (A=triangle, B=circle, C=square, D=star), message strip ABACABDA, pencil, and optional movable icon labels. A paper divider can hide the sender's original message.

Preparation: about 8 minutes. 0-10 send short strings with the fixed-length code; 10-20 discover ambiguity; 20-35 design a shorter prefix code; 35-40 walk left/right tree branches; 40-55 prove optimality or change frequencies; 55-60 decode a partner's test string.

Stop: A working 14-bit prefix code and either its decoding explanation or its complete two-shape optimality proof. The frequency reversal and six-message tie are worthwhile extensions, not filler.

Staged hints

1. For 29.2: Try a cut after the first bit. Then try a cut after the second.
2. For 29.3: The four pictures are not equally common. Try giving the most common one a single-edge leaf, and put the other three under the other branch.
3. For 29.5-29.6: Count leaves on the two sides of the root. For labels, compare the cost before and after swapping a frequent deep picture with a rare shallow one.

Checked solutions to every prompt

29.1

ABACABDA encodes as 00|01|00|10|00|01|11|00, transmitted 0001001000011100:16 bits. Because all words have length 2, pairwise decoding is unambiguous. The printed icon message includes fourA, twoB, oneC, oneD.

Send it with fewer bits

29.2

010=0|10 gives AC, while 010=01|0 gives BA. This specific complete proposed code is genuinely ambiguous. A is a prefix of B, and a greedy stop at 0 can prevent decoding 01 as B. Do not generalize that every non-prefix code must be ambiguous; the worksheet restricts to leaf/prefix codes to guarantee immediate decoding.

29.3

One successful code is A=0, B=10, C=110, D=111, using 14 bits. The original strip encodes 0|10|0|110|0|10|111|0, transmitted 01001100101110. Many mirrored codes and a C/D swap work equally well. A valid code must give four distinct nonempty leaf words with no word a prefix of another. An impressive bit total achieved with hidden delimiters is not valid.

29.4

Suppress a node with exactly one child by bypassing that edge/subtree step; every descendant code shortens by 1, other code lengths do not increase, and the leaf relation still gives a prefix code. All frequencies are positive, so the total strictly decreases. Root-unary steps can also be removed. Therefore a shortest prefix tree has no unary internal nodes. The tiny picture illustrates the operation; it does not claim there are only two messages in the main problem.

29.5

At the root the positive leaf counts split 1+3 or 2+2, up to exchanging sides. A two-leaf full tree is a fork. A three-leaf full tree splits 1+2. Thus the only four-leaf depth patterns are 1, 2, 3, 3 and 2, 2, 2, 2. Planar mirror variations add codes but not new length shapes. Best skew labeling puts A at depth 1, B at 2, C/D at 3; balanced labeling is arbitrary.

29.6

Balanced total $2(4+2+1+1)=16$. Skew total $4 \times 1 + 2 \times 2 + 1 \times 3 + 1 \times 3 = 14$. Exchange proof: if a frequent picture has a longer path than a less frequent one, swapping them saves one bit per extra occurrence per exchanged depth step, and cannot increase cost. For example, weights 4 and 1 at depths 3 and 1 cost 13; swapped cost 7. Sorting larger counts toward shorter depths minimizes each shape. Removing unary steps and classifying both full shapes exhausts all prefix-code possibilities, so 14 is optimal among binary prefix codes.

29.7

All counts are 2. Balanced total $8 \times 2 = 16$. Skew total $2(1+2+3+3)=18$. The balanced tree wins; yesterday's skew winner now loses. A rarer message can be longer in a code designed to minimize total length, so one cannot optimize by asking only for the longest codeword.

Send it with fewer bits

29.8

Use frequencies 2, 2, 1, 1, for example ABACBD. A balanced code costs $6 \times 2 = 12$. A skew code placing one doubled picture at depth 1 and the other at 2 costs $2 \times 1 + 2 \times 2 + 1 \times 3 + 1 \times 3 = 12$. All placements of the two doubled labels in the first two depths tie. With six total messages and all four present, the only frequency multisets are 3, 1, 1, 1 (skew 11 vs balanced 12) and 2, 2, 1, 1 (tie 12); hence this also characterizes all possible successful frequency multisets.

29.9

Any correctly decoded partner message using a verified prefix tree is acceptable. Starting at the root, each bit chooses exactly one edge. A symbol is emitted only at a leaf, so no longer codeword could share that entire word as a proper prefix. Return to the root and repeat. Induction over symbols shows every concatenation decodes uniquely. A partial final word or traversal off the tree flags an invalid/incomplete transmission; it is not an alternate valid decoding.

What this really teaches

Finite lossless source coding; weighted external path length; tree reduction and exchange arguments. This is a complete small optimality proof without giving Huffman's algorithm as a recipe. The general continuation asks why two least-frequent symbols can be placed as deepest siblings and merged recursively. Entropy/logarithms are unnecessary for this finite task.

Prior use: Week 6's code-query trees optimize questions to identify an object. Here one sends a sequence with no boundaries, and weighted communication cost is the objective. Both use trees; the constraints and the proof are different.

Source adaptation: AP-29's Shannon source supplies the code framework and the classical 4, 2, 1, 1 example. The sender/receiver script, ambiguous strip, classification prompts and changed-frequency challenges are instructional adaptations, checked directly.

Claude Shannon, A Mathematical Theory of Communication. Part I, PDF p. 18: code 0,10,110,111 and weighted average length.

<https://people.math.harvard.edu/~ctm/home/text/others/shannon/entropy/entropy.pdf> Evidence status: relevant source framework inspected; finite example explicitly proved below

Verification: [plans/atlas/worksheet-trial/decisions-checks.py](#); exact enumeration and rational calculations in [decisions-checks-results.json](#). Full battery state tables are in that checked data; they are deliberately absent from the student work.